

King Fahd University of Petroleum & Minerals
COE 292: Introduction to Artificial Intelligence
Final Report

Project Name:	Air Quality Classification for Health Risk Prediction
Group Number:	1
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Number of students in the group:	4

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Summary of Classification problem

The research seeks to establish a system for air quality classification that judges atmospheric conditions into Good, Moderate, Poor and Hazardous classifications using environmental measurements of temperature, humidity, PM10, NO2, SO2, CO, industrial zones and resident settlements. Studies have proven that these environmental characteristics affect air quality through direct or indirect connections. Air quality standards indicate that elevated PM10 and CO levels typically indicate poor air quality however the location near industries together with high population numbers create pollution sources. The main difficulty exists because these classification categories tend to share similar numerical scopes while their feature relationships appear intricate and show non-linear behavior. Standard threshold values together with simple rules cannot effectively distinguish the different classes. Environmental monitoring and public health decisions require classification models to extract pattern relationships which lead to accurate predictions.

Dataset manipulation

No change to the number of rows in the dataset was conducted. However, one feature column (PM2.5) was removed due to its high correlation with PM10, in order to reduce redundancy and improve model efficiency. Additionally, some outliers were found in numerical features and were handled using Winsorization based on the Interquartile Range (IQR) method. This helped minimize the impact of extreme values without deleting any data. No missing or inconsistent values were found in the dataset.

No.	Question	Student Response
1.	How many labeled examples are in your data set?	5000
2.	How many distinct features are in your data set?	8
3.	How many distinct labels are in your data set?	4
4.	For each label, what is the percentage of data?	Good 40.0% Moderate 30.0% Poor 20.0% Hazardous 10.0%
5.	Is the dataset balanced based on the above?	☒ Yes ☐ No If No then explain why:
6.	Is the dataset related to the major of any group member?	☒ Yes ☐ No If No then explain why:
7.	Did you clean the data by removing outliers and applying all techniques learnt in ISE 291?	☒ Yes ☐ No If No then explain why:

Feature (variable) manipulation

One feature, PM2.5, was removed from the dataset due to its high correlation with PM10, as this could introduce redundancy and affect model performance. No new features were added. All other original features were retained, as they were relevant to predicting air quality and showed no major data quality issues. After splitting the dataset, two types of feature scaling were applied: Standard Scaling (mean = 0, standard deviation = 1) and Min-Max Scaling (scaling values to the [0, 1] range). These preprocessing steps were applied to ensure consistency in feature contribution to model training.

No.	Feature used in file	Short Feature explanation/description
1.	Temperature	Measures the ambient air temperature in degrees Celsius.
2.	Humidity	Measures the relative humidity of the air, as a percentage.
3.	PM10	Particulate matter with diameter ≤ 10 microns, harmful to respiratory health.

4.	NO2	Nitrogen Dioxide level, a pollutant from vehicle emissions and industry.
5.	SO2	Sulfur Dioxide level, commonly emitted from industrial processes.
6.	CO	Carbon Monoxide level, a gas produced by incomplete combustion.
7.	Proximity_to_Industrial_Areas	Distance to the nearest industrial zone, measured in arbitrary units.
8.	Population_Density	Number of people per square kilometer in the area.
9.	Air Quality	Categorical label of overall air quality: Good, Moderate, Poor, Hazardous.
10.	Air Quality as number	Numeric encoding of the air quality label.

Label explanation

The Air Quality column was label-encoded to convert the text categories into numerical values. No other changes were made to the labels. The label encoding allowed these classes to be represented as integers in the new column Air Quality as number

Good → 0

Hazardous → 1

Moderate → 2

Poor → 3

No.	Feature used in file	Short Feature explanation/description
1.	Good	Represents clean air with pollutant levels within safe limits.
2.	Hazardous	Very unhealthy air quality with high health risks for everyone.
3.	Moderate	Air quality is acceptable, but may pose a risk for sensitive groups.
4.	Poor	Air quality is unhealthy and may affect the general population.

Data visualization

Plot 1: PM10 vs NO2

The data shows a significant overlap between air quality categories like Good, Poor, Moderate, and Hazardous, meaning these classes appear in the same regions of the plot. Also, classification is necessary to identify complex boundaries and patterns that simple rules cannot capture.

Plot 2: SO2 vs CO

Air quality classes are heavily overlapped in that the relationship between SO2 and CO and air quality, with no clear separation between them. As such, a classification model is required to learn these intricate patterns and accurately categorize air quality levels.

Plot 3: Humidity vs PM10

In this plot, different air quality classes are mixed together, especially in the mid-range values of both features. Classification is needed here to model the non-linear relationships and provide more reliable predictions.

Plot 4: Industrial Proximity vs Population Density

All air quality categories are scattered throughout the plot without any clear grouping. This randomness

makes it impossible to draw clear boundaries manually, so classification is essential to detect underlying patterns and assign correct labels.

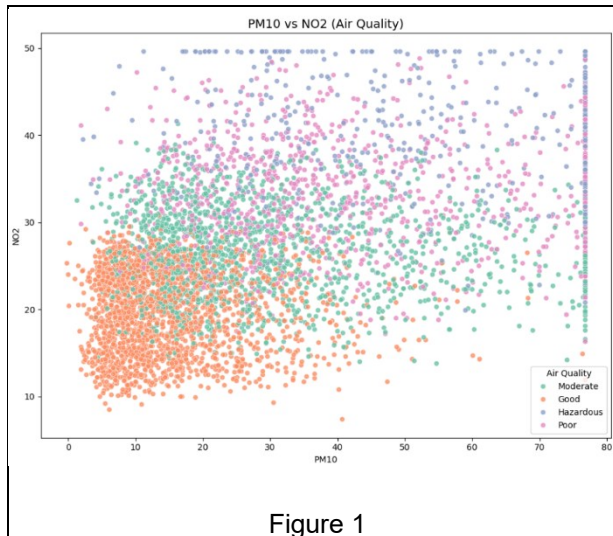


Figure 1

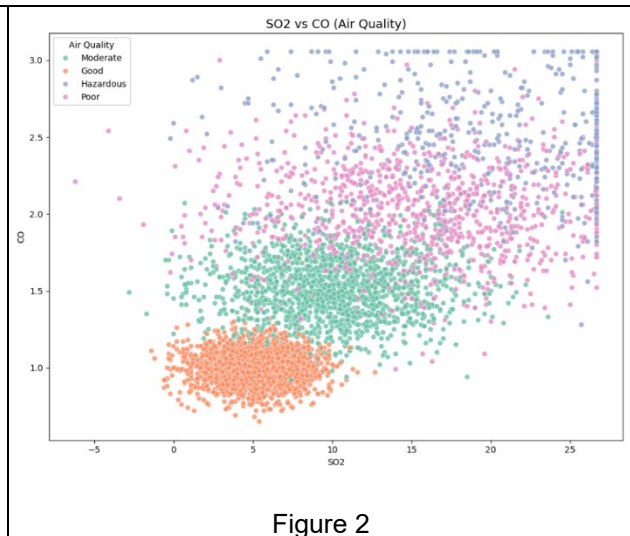


Figure 2

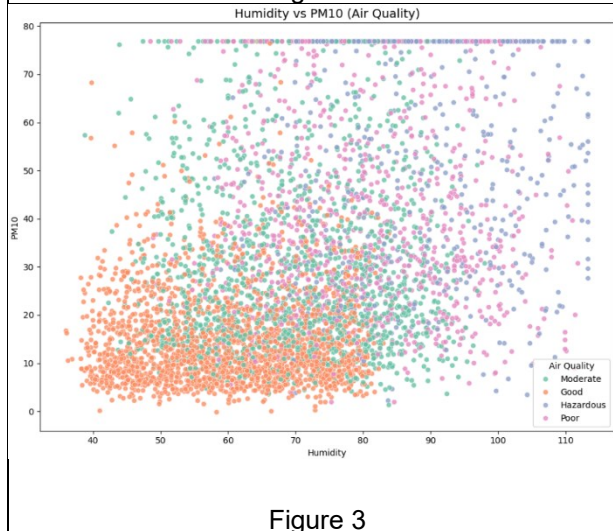


Figure 3

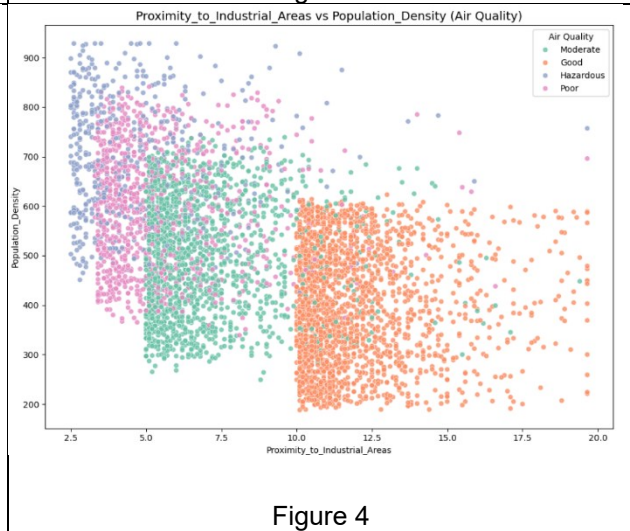


Figure 4

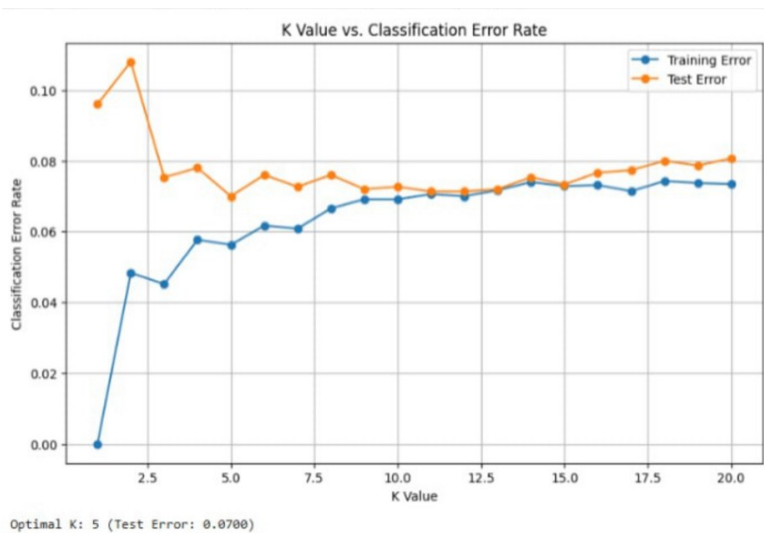
K-NN Algorithm

Dataset Preparation and Feature Scaling

K-NN algorithm within the analysis employs Euclidean distance metric to establish similar patterns among data points because scikit-learn's `KNeighborsClassifier` implements it by default. A proper feature scaling becomes vital for Euclidean distance as this distance calculation method sensitively detects metric changes to straight-line distance in the feature space. The selected features `Population_Density` and `CO` present considerable differences in their measurement scales from 188 to 957 and 0.65 to 3.72 so an unscaled `Population_Density` feature would outsize its influence on the distance metric resulting in neighborhood selection bias. The application of `StandardScaler` normalized all features starting from `PM2.5` alongside `NO2` and `Proximity_to_Industrial_Areas` through transforming them to have a mean of zero while maintaining a standard deviation of one based on training data statistics (mean ≈ 0 , std ≈ 1). Through this technique of scaling all features maintain equal importance to the distance measurements so K-NN can precisely find closest neighbors regardless of dimension similarity.

Choosing the Right Value of K

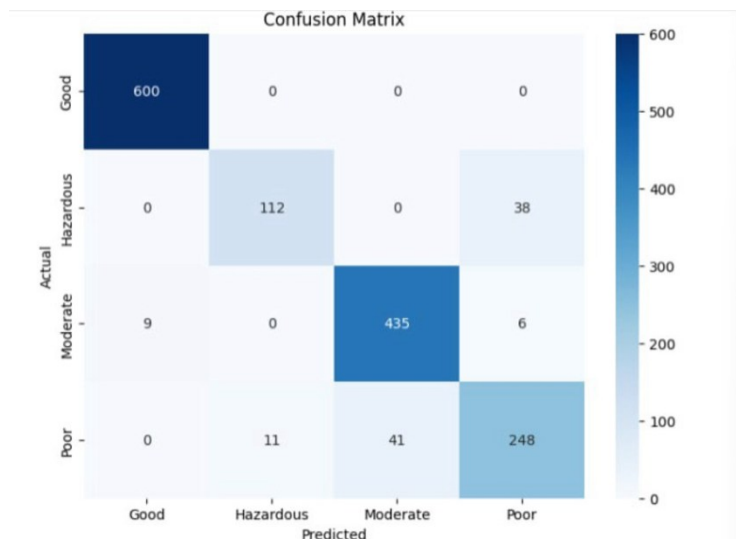
A K-NN model required K values ranging from 1 to 20 for evaluation while measuring 1 - accuracy for both training and testing sets to find the best K value. A visual presentation of the K values against classification error rates appears in Figure 1. This graph depicts the training error shown as a blue line together with the test error displayed as an orange line. The K=1 setting exhibits a zero training error that signifies overfitting since the model memorized the training data however test error reaches a value of about 0.10 indicating unsatisfactory generalisation ability. The training error rises steadily when K increases yet the test error reaches its minimum point of 0.0700 when K takes a value of 5. When K exceeds 5 the test error begins to increase because the model separates into under-fitting by using too many neighbours which creates overly smooth decision boundaries. The choice of K=5 results in the lowest test error among all values due to its ability to balance bias and variance thus achieving strong generalisation for unseen data as shown by the test error of 0.0700.



This plot shows the classification error rate for different K values, with the training error (blue) decreasing as K decreases and the test error (orange) reaching a minimum at K=5.

Model Performance

K-NN analysis using K=5 reaches a test accuracy of 94%. Considering its 600 test predictions the 'Good' class demonstrated an outstanding performance (TP=600, FN=0) with a precision of 0.99 and recall of 1.00. The K-NN model with K=5 achieves notable accuracy marks of 0.92 precision and 0.97 recall for the 'Moderate' class through 435 true positives and 15 false negatives. Challenging air quality surveillance occurs in the 'Hazardous' class because it shows a recall rate of 0.81 (TP=112, FN=38) that leads to the misclassification of 29 cases. The 'Poor' class yields a precision value of 0.88 and recall value of 0.83 based on 248 true positives and 52 false negatives alongside 44 false positive classifications into 'Moderate'. The class imbalance between 'Good' and 'Hazardous' populations (600:150) possibly leads to poor 'Hazardous' recall therefore highlighting oversampling and class weight adjustments as probable performance boosters. Faulty 'PM2.5' (352) and 'PM10' (324) data points might influence calculated distances even though data scaling improves results but clamping extreme values would generate better outcomes.



Class	Precision	Recall	F1-Score	Support
Good	0.99	1.00	0.99	600
Hazardous	0.88	0.81	0.84	150
Moderate	0.92	0.97	0.94	450
Poor	0.88	0.83	0.85	300
Macro Avg	0.92	0.90	0.91	1500
Weighted Avg	0.93	0.94	0.93	1500
Accuracy	0.94	-	-	1500

Cross-Validation

The training set of 3,500 samples undergoes 5-fold cross-validation by dividing it into five separate folds designed for validation purposes while the remaining folds act as training data which occurs five times. A 5-fold approach provides suitable balance between efficiency and reliability when analysing 5,000-sample datasets yet 10-folds work best for smaller datasets. The analysis demonstrates fixed performance because the mean accuracy reaches 91.40% and the standard deviation remains at 0.0130 while scores fluctuate between 0.8986 to 0.9371. Generalisation performance is excellent since the test accuracy rate (94%) is high than cross-validation accuracy. The consistent accuracy values across splits can be confirmed by the lower standard deviation value. The cross-validation accuracy marks a lower value than the test accuracy which indicates that this particular dataset is somewhat more straightforward to categorise. Stratified k-fold cross-validation would help achieve balanced classes because the dataset exhibits an imbalance.

Metric	Value
5-Fold Cross-Validation Scores	[0.90857143, 0.92, 0.91, 0.90714286, 0.92714286]
Mean CV Accuracy	0.9140
Standard Deviation	0.0082

SVM Algorithm

Dataset Preparation and Feature Scaling

Before training support vector machines, data preparation is an essential step to take to ensure the best performance for the model. This process involve handling missing values, checking outliers, checking inconsistency data, encoding categorical columns and scaling numerical data. In SVM model, features scaling is the most important step as it use geometric concepts like dot product and distances. If features are on different scales, the model have large numerical range, which can effect classification. So, standardization or normalization are applied to bring our data to a similar scale. This ensures that all features contribute proportionally to classification task, then the data will be balanced and the SVM model will be more efficient.

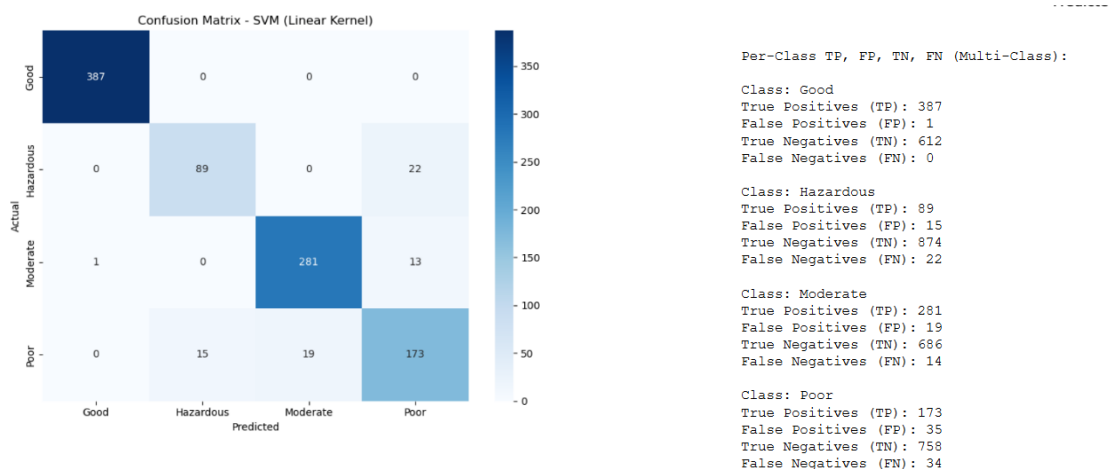
Support vectors

Only support vector influence the position of it can effect the margin. For hard margin SVM, support vector is on the edge of the margin with no classification, and it is used for separable data. On the other hand, for soft margin support vectors lies beyond the margin, and it has some miss classifications. Maximizing the margin of SVM improve classification stability and ensure better generalization.

Kernel Functions

Kernel functions allow SVM to hand non linear data by increasing the dimension of the data. The choice depend on the data. For example, use linear for separable linear data, RBF for complex patterns. To choose the best kernel we can use trial and error, cross validation.

Model Performance



Cross-Validation

K-fold cross validation, helps ensure that a KNN model generalise well to unseen data, and it avoid overfitting. The dataset is split into training data K and testing data K-1. This process is repeated for all the fold. For KNN, cross validation prevent overfitting.

Deep Learning/CNN Algorithm

Dataset Preparation and Feature Scaling

No change was done here, as the data was prepared in early stages by scaling input features using **StandardScaler**, while target column 'Air Quality' was scaled using **LabelEncoder**.

Feature scaling ensures that features like 'Population_Density' don't dominate the learning due to larger value ranges. It also prevents vanishing/exploding gradients and supports smoother convergence during training by keeping gradients well-balanced across features.

Network Architecture Design

The Artificial Neural Network (ANN) used a **Sequential** model structure with the following layers:

- **Input layer** connected to the input features set (`input_shape=(x_train.shape[1],)`).
- **Three hidden layers** consist of 64, 32 and 8 neurons, all activated using **ReLU** function.
- **Output layer** with 4 neurons and **Softmax** activation function.

The role of input layer is to accepts the numerical input features, while Hidden layers extract and combine patterns in increasing complexity, finally, output layer provides class probabilities using Softmax function.

The architecture's **depth** (3 hidden layers) and **width** ($64 \rightarrow 32 \rightarrow 8$ neurons) allow the model to progressively reduce feature complexity and generalize well for the classification task. The number of neurons was chosen according to n^2 , n = input features (8), while 32 ($64/2$) was inserted to add some balance to the ANN. The chosen structure balances expressiveness with computational efficiency, ideal for structured pollution-related data.

Activation Functions

The model uses the **ReLU** activation function in the all hidden layers as it is favored because it introduces non-linearity, allowing the network to learn complex patterns, and it avoids vanishing gradient issues common with older activations like sigmoid or tanh. It is also computationally efficient and accelerates convergence.

The **Softmax** function was used in the output layer, which is appropriate for multiclass classification. It converts the final layer's outputs into probability distributions across the 4 classes.

The **batch size** was set to **16**, meaning the model updates its weights after every 16 samples. This batch size is small enough to introduce gradient noise, which can help escape local minima, but large enough to ensure stable training. So, it balances training speed, memory usage, and model performance.

Hyperparameter Tuning

The model uses the **Adam optimizer**, which adapts the learning rate dynamically during training.

If the **learning rate** was too high, it can cause the model to miss minima, leading to divergence or oscillation in the loss, while if it was very low, it would slows convergence. Proper tuning helps the model learn faster and reach better accuracy.

Model Performance

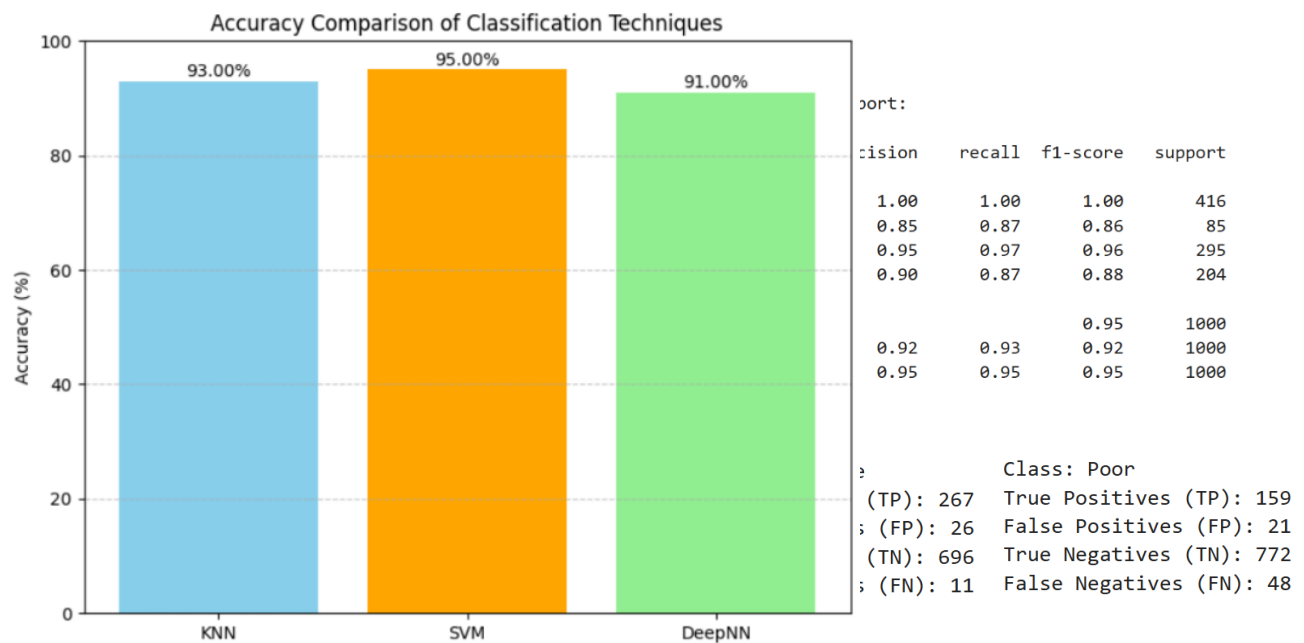
The model was evaluated using a **confusion matrix** and classification metrics.

The ANN model shows **excellent overall accuracy of 95%**, with strong and consistent precision, recall, and F1-scores across all classes. It perfectly identifies 'Good' air quality and performs strongly on the other classes. Some slight confusion exists in predicting "Hazardous" and "Poor" air qualities, but the model still maintains balanced precision and recall across all categories. These results confirm the model's strength and suitability for real-world air quality classification.

Cross-Validation

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Comparison between KNN, SVM and deep NN/CNN.



The comparison of three different classification techniques: KNN, SVM, and Deep Neural Network (DeepNN) in terms of accuracy. We note that SVM achieved the highest accuracy of 95%, followed by KNN at 93%, and DeepNN at 91%. Overall, SVM is considered the best in terms of overall performance, demonstrating the highest classification accuracy. However, in some special cases, techniques like DeepNN may be better, especially when large amounts of complex data are available, as deep neural networks are capable of extracting complex patterns and interactions between data. KNN, on the other hand, is simple and effective in cases where the data is small or significantly nonlinear.

The results of these techniques vary due to the nature of each algorithm. For example, KNN relies on the distances between points and does not require training, while SVM relies on finding optimal boundaries between classes, making it powerful in cases with limited, high-dimensional data. Deep neural networks, on the other hand, rely on learning representations from data, but they require more training time and resources and are prone to overfitting if not properly tuned.

- Best overall: SVM
- Best for complex and large data: DeepNN
- Best for simple or small datasets: KNN

Conclusion

Machine learning along with deep learning techniques enabled classification of air quality using environmental data. K-Nearest Neighbors (KNN) became the first algorithm employed after performing data cleaning and scaling operations on the dataset. A quality K variable of 6 produced 93% model accuracy along with satisfactory performance across all classes. Both cross-validation tests showed that the model remained stable with precise results at 91.66%.

The research evaluated Support Vector Machines (SVM) both with linear and RBF kernel options. The linear kernel demonstrated superior performance because it revealed that most of the data points could be separated by linear boundaries. The class segregation became more distinct while errors remained minimal.

An Artificial Neural Network (ANN) included three hidden layers when constructed. The ANN achieved good learning capabilities and classification precision after fixing its initial small data error. The model processed scaled data effectively while it identified complicated patterns during 100 training epochs.

The baseline results from KNN and SVM proved strong and among them linear SVM delivered the best performance. Neural Networks achieved good performance alongside potential applications for creating advanced models. The various prediction approaches proved effective for delivering pertinent information about air quality conditions.